

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7.	Indicative content Section A: Environmental impacts • (Excess) use of (inorganic) fertiliser • {Nitrates/ fertilizer} {washed/ leached/ run off} into {river/ water} • Causing algal blooms {along banks of Mississippi/ shown in image 7.1} • Blocking sunlight {causing plants to die/ means plants cannot photosynthesise} • Bacteria decompose plants using up oxygen • causing {lack of oxygen environment and {fish/ organisms} to die/ dead zones in {image 7.2/ named location} Section B: Root nodules • GM plants grew much better in low nitrate • Root nodules contain {N fixing bacteria/ rhizobium} • Correct reference to symbiosis/ mutualism • ({Rhizobium/ bacteria} in root nodules was) providing {nitrogen containing compounds/ nitrates/ ammonium} Section C: Lessening impact • Use of node forming plants increases {N content of soil/ soil fertility} • {Less/ no} (inorganic) fertiliser needed • Less {algal blooms/dead zones/ eutrophication} • Higher yield therefore less land needed for crops • Therefore more land available {for habitats/ increasing biodiversity}						